

Review Session of 2nd Week *

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Contents

1	Difference Equations	2
1.1	First-order Difference Equation	2
1.2	Second-Order Difference Equations	2
2	A Typical Dynamic Problem	3
2.1	Kuhn-Tucker Conditions	3
2.2	Social Planner's Problem	4
2.3	Decentralized Economy	5
2.3.1	Households	5
2.3.2	Firms	6
2.3.3	General Equilibrium	6

*Most of the materials follow the lecture notes of Professor Zhang.

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1 Difference Equations

1.1 First-order Difference Equation

A first-order difference equation in x_t can usually be written in the form

$$x_{t+1} = f(t, x_t), \quad t = 0, 1, 2, \dots \quad (1)$$

where $f(t, x_t)$ is a function: $N \times R \rightarrow R$.

Example.1 Homogeneous Difference Equation

$$x_{t+1} = ax_t, \quad t = 0, 1, \dots \quad (2)$$

where a is a constant.

Suppose x_0 is given, the solution for this *linear*¹ difference equation is

$$x_t = a^t x_0, \quad t = 0, 1, \dots \quad (3)$$

Example.2 Nonhomogeneous Difference Equation

$$x_{t+1} = ax_t + b, \quad a \neq 1, \quad t = 0, 1, \dots \quad (4)$$

A simple and efficient way to solve this equation is to transform nonhomogeneous difference equation into homogeneous case using the following undetermined coefficients method. Define $z_t \equiv x_t + \alpha$ with α as the undetermined coefficient. The new difference equation is $z_{t+1} = az_t$. In order to make sure the two equations are equivalent, we require the following relation to be true

$$x_{t+1} + \alpha = a(x_t + \alpha) \iff x_{t+1} = ax_t + b$$

Thus we know $(1 - a)\alpha = b$, and $\alpha = -\frac{b}{1-a}$. So we can write down the solution easily as

$$x_t = a^t \left(x_0 - \frac{b}{1-a} \right) + \frac{b}{1-a} \quad (5)$$

1.2 Second-Order Difference Equations

A typical second-order linear difference equation takes the form

$$x_{t+2} + a_1 x_{t+1} + a_2 x_t = c \quad (6)$$

Again, we firstly consider the homogeneous case to see what we can get. Actually, we have the following theorem.

Theorem The general solution of

$$x_{t+2} + a_1 x_{t+1} + a_2 x_t = 0 \quad a_2 \neq 0$$

is as follows:

¹Hereafter, we only consider the linear case.

1. If $a_1^2 - 4a_2 > 0$ (the characteristic equation has two distinct real roots),

$$x_t = A\gamma_1^t + B\gamma_2^t \quad \gamma_{1,2} = -\frac{1}{2}a_1 \pm \frac{1}{2}\sqrt{a_1^2 - 4a_2}$$

2. If $a_1^2 - 4a_2 = 0$ (the characteristic equation has one real root),

$$x_t = (A + Bt)\gamma^t, \quad \gamma = -\frac{1}{2}a_1$$

3. If $a_1^2 - 4a_2 < 0$ (the characteristic equation has no real roots),

$$x_t = r^t(A \cos \theta t + B \sin \theta t), \quad r = \sqrt{a_2}, \cos \theta = -\frac{a_1}{2\sqrt{a_2}}, \theta \in [0, \pi]$$

Theorem The solution for a second-order nonhomogeneous linear difference equation with constant coefficients, x_t^* , can be described as the summation of the solution of its associated homogeneous equation x_t and its particular solution $\bar{x}_t$.

$$x_t^* = x_t + \bar{x}_t \quad (7)$$

Example.3 Nonhomogeneous Second-Order Difference Equations

$$x_{t+2} + 6x_{t+1} + 9x_t = 16 \quad (8)$$

The solution of this equation is $x_t = A(-3)^t + Bt(-3)^t + 1$, where A and B can be assigned definite values, given two initial conditions.

2 A Typical Dynamic Problem

2.1 Kuhn-Tucker Conditions

A standard nonlinear programming problem is

$$\max f(x_1, \dots, x_N) \text{ s.t. } \begin{cases} g_1(x_1, \dots, x_N) = b_1 \\ \dots \\ g_M(x_1, \dots, x_N) = b_M \\ h_1(x_1, \dots, x_N) \leq c_1 \\ \dots \\ h_K(x_1, \dots, x_N) \leq c_K \end{cases} \quad (9)$$

Theorem (*Kuhn-Tucker Conditions*) Suppose that $x^* = (x_1^*, \dots, x_n^*)$ solves problem (9) where f and $g_1, \dots, g_m$ are C^1 functions. Assume that the constraint qualification² is satisfied. Then there are multipliers $\lambda_m \in R$, with $m = 1, \dots, M$, and $\lambda_k \in R_+$, with $k = 1, \dots, K$, such that:

²The constraints are independent at x^* .

1. For every $n = 1, \dots, N$,

$$\frac{\partial f(x^*)}{\partial x_n} = \sum_{m=1}^M \lambda_m \frac{\partial g_m(x^*)}{\partial x_n} + \sum_{k=1}^K \lambda_k \frac{\partial h_k(x^*)}{\partial x_n}$$

2. For every $k = 1, \dots, K$,

$$\lambda_k (h_k(x^*) - c_k) = 0$$

which are also called *Complementary Slackness* conditions.

2.2 Social Planner's Problem

The optimization problem

$$\max \sum_{t=0}^T \beta^t u(c_t) \quad (10)$$

s.t.

$$c_t + k_{t+1} \leq f(k_t) + (1 - \delta)k_t$$

$$c_t \geq 0, t = 0, 1, \dots, T$$

$$k_{t+1} \geq 0, t = 0, 1, \dots, T$$

$$k_0 > 0 \text{ is given.}$$

The assumptions about the utility function and production function are the same as those in your lectures.

Assumption 1 $u(\cdot)$ is strictly increasing, differentiable, concave and $f(\cdot)$ is strictly increasing, differentiable, concave with $F(0, L) = F(K, 0) = 0$.

So the problem can be rewritten as

$$\max \sum_{t=0}^T \beta^t u(f(k_t) - k_{t+1}) \quad (11)$$

s.t.

$$0 \leq k_{t+1} \leq f(k_t), t = 0, 1, \dots, T$$

$$k_0 > 0 \text{ is given.}$$

Assumption 2 (*Inada condition on utility function*) $\lim_{c \rightarrow 0} u'(c) = \infty$.

Lemma 1 Under the Inada condition on utility, the optimal solution is interior, that is, $k_{T+1} = 0$ and $0 < k_{t+1} < f(k_t)$ for all $t = 0, 1, \dots, T-1$.

Proof

Suppose $\bar{k} \equiv (\bar{k}_1, \bar{k}_2, \dots, \bar{k}_t, \bar{k}_{t+1}, \dots, \bar{k}_T, \bar{k}_{T+1})$ solves the optimization problem (11). Without loss of generality, let denote $x \equiv \bar{k}_{t+1}, t = 0, 1, \dots, T-1$, which should satisfy the following optimal conditions:

$$\begin{cases} -\beta^t u'(f(k_t) - x) + \beta^{t+1} u'(f(x) - k_{t+2}) f'(x) = 0, & \text{if } 0 < x < f(k_t) \\ -\beta^t u'(f(k_t) - x) + \beta^{t+1} u'(f(x) - k_{t+2}) f'(x) \leq 0, & \text{if } x = 0 \\ -\beta^t u'(f(k_t) - x) + \beta^{t+1} u'(f(x) - k_{t+2}) f'(x) \geq 0, & \text{if } x = f(k_t) \end{cases}$$

- If $x = 0$, $f(x)$ will be zero as well. Thus, $f(x) - k_{t+2}$ must be zero. Given the Inada condition on utility, we know that $u'(f(x) - k_{t+2}) \rightarrow \infty$. So $-\beta^t u'(f(k_t) - x) + \beta^{t+1} u'(f(x) - k_{t+2}) f'(x) > 0$. There is a contradiction, which means $x \neq 0$.
- If $x = f(k_t)$, $c_t = f(k_t) - x = 0$. With the same reason, we get $u'(f(k_t) - x) \rightarrow \infty$. Thus $-\beta^t u'(f(k_t) - x) + \beta^{t+1} u'(f(x) - k_{t+2}) f'(x) < 0$. Again, we have the similar conclusion that x can not be $x = f(k_t)$.

For $t=T$, we get $-\beta^T u'(f(k_T) - k_{T+1}) \leq 0$, thus k_{T+1} should be zero.

To sum up, we have the result that $k_{T+1} = 0$ and $0 < k_{t+1} < f(k_t)$ for all $t = 0, 1, \dots, T-1$.

Q.E.D.

Thus we get the first-order conditions as Euler equation

$$u'[c_t] = \beta u'[c_{t+1}] f'(k_{t+1}), \quad t = 0, 1, \dots, T-1 \quad (12)$$

The economic meaning for this inter-temporal condition is that the following two activities will produce the same marginal utility: to increase your consumption today, to invest your additional saving and consume the return tomorrow.

2.3 Decentralized Economy

There is no technology progress.

2.3.1 Households

- L amounts of households live an infinite amount of time constantly.
- Each household supplies one unit of labor inelastically.
- Households earn income by working and investing. So household's wealth at the beginning of each period is $a_t = R_t k_t + w_t + (1 - \delta)k_t$.
- Households make the decisions of consumption and saving optimally.

Thus household's problem is

$$\max \sum_{t=0}^T \beta^t u(c_t) \quad (13)$$

s.t.

$$c_t + k_{t+1} \leq R_t k_t + w_t + (1 - \delta)k_t$$

$$c_t \geq 0, t = 0, 1, \dots, T$$

$$k_{t+1} \geq 0, t = 0, 1, \dots, T$$

$$k_0 > 0 \text{ is given.}$$

2.3.2 Firms

- Firms employ labor and rent capital in competitive labor and capital markets to produce the final good.
- The technology for each firm j is $Y_j(t) = F(K_j(t), L_j(t))$.
- The firms seek to maximize profits. The FOCs for an interior solution require

$$F_K(K_j(t), L_j(t)) = R_t$$

$$F_L(K_j(t), L_j(t)) = w_t$$

2.3.3 General Equilibrium

- Market Clearing:

$$\int_j K_j(t) dj = K(t) \text{ and } \int_j L_j(t) dj = L$$

- Given the market structure, we know

$$R_j(t) = R_i(t) = f'(k_t), \text{ and } w_j(t) = w_i(t) = f(k_t) - f'(k_t)k_t$$

Thus the household's problem is equivalent to the social planner's problem.

Theorem 1 (*First welfare theorem*) The allocation in a competitive Equilibrium solves the social planners problem (it is pareto efficient).

Theorem 2 (*Second welfare theorem*) If an allocation is pareto efficient, then we can find prices such that the prices and the allocation constitute an equilibrium.